

Amirreza Radjou

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years building backend services and distributed systems. Strong in Go and Python across microservices, gRPC/REST/GraphQL APIs, service mesh, and consensus/peer-to-peer protocols. Proven in regulated banking and research settings, with solid testing, CI/CD, and containerized delivery; comfortable full-stack with Vue.js and React.

TECHNICAL SKILLS

Languages: Go, Python, TypeScript, JavaScript, Java, Rust, SQL

Backend & APIs: gRPC, REST, GraphQL, Django, Flask

Distributed Systems: Microservices, service mesh, peer-to-peer (IPFS), consensus & blockchain protocols (Substrate/Polkadot)

DevOps & Platforms: Docker, Kubernetes, OpenShift, Jenkins, GitHub Actions, CI/CD

Data Stores: PostgreSQL, ETL/ELT pipelines, custom data loaders

Frontend: Vue.js, React, TypeScript, HTML5/CSS3

Testing: JUnit, pytest, Cypress (e2e)

Security: API security & mTLS, threat modeling, penetration testing, cryptography

PROFESSIONAL EXPERIENCE

CIBC

Senior Full Stack Developer

Toronto, ON

March 2026 – Present

- Hardened REST and gRPC endpoint security by deploying a **service mesh on OpenShift** (mTLS, traffic policies, observability) for zero-trust communication between microservices.
- Added **gRPC** APIs for inter-service communication, cutting payload overhead and latency versus REST on high-throughput internal traffic.
- Onboarded new product models — including **Simplii Financial** — into the Enterprise Product Catalog (EPC), the authoritative product-data source business teams depend on.
- Contributed to architecture and design decisions enabling shared services to serve multiple customer brands.
- Authored and presented Interface Agreements and provided integration support to partner teams.

Dandelion Network

Software Engineer

Remote

May 2024 – March 2025

- Built **Go and Python** backend services for a mobile wallet, exposing REST, **GraphQL** (chat, transaction history), and **gRPC** inter-node communication across a distributed network.
- Led the network simulator team, building an environment to evaluate node and client behaviour under adversarial conditions — malicious nodes, weak connectivity, and network delays.
- Developed Substrate/Ink! (**Rust**) smart contracts for multi-signature wallets and a secure Vault on Polkadot, hardened against reentrancy, with a Vue.js + TypeScript frontend.
- Drove security testing via simulated attacks; standardized Jenkins CI/CD and Docker for repeatable releases.

York University

Research Assistant – Software Engineer

Toronto, ON

September 2022 – May 2025

- Extended CNSim, an open-source Java network simulator, with Bitcoin and IOTA/Tangle protocols for comparative analysis of distributed consensus systems.
- Introduced a Transaction Finality metric and pluggable node-behaviour strategies to benchmark protocols across settings and attack scenarios.
- Applied design patterns for extensibility, added JUnit test harnesses, and built Python post-processing scripts and a visualization GUI for network behaviour.

SELECTED PROJECTS

CO₂ Emissions in Bitcoin Mining — Full Stack Developer

Academic, Sep 2023 – Apr 2024

- Built a full-stack app (React/TypeScript, Django/Flask, **PostgreSQL**) visualizing Bitcoin-mining CO₂ emissions as an interactive, time-ranged geospatial heatmap.
- Optimized in-browser rendering of large datasets via chunked delivery and precomputed cumulative aggregates; automated a daily multi-source ETL pipeline feeding a REST API.

go-orbitdb

Go implementation of OrbitDB (peer-to-peer database on IPFS) with key-value, event-log, and document stores, replication for eventual consistency, and cryptographic access control.

Open Source

EDUCATION

M.Sc. Computer Science, York University, Toronto, ON

Sep 2022 – Oct 2025

B.Sc. Computer Science, Amirkabir University of Technology, Tehran

Oct 2018 – Aug 2022